Operating Systems (OS)

MCA 1st Year, 1st Semester

Prepared By-
Mr. Asheesh Pandey
(MCA, MTech(CS), PhD(CS))
Assistant Professor (Sr. Scale)
Dept. of Computer Applications
ABES Engineering College UP.

Unit-1
a. Introduction

Lecture-1

Operating System-

- ⇨ A system program that acts as an intermediary between computer user and the computer hardware.
- ⇨ It is an important part of almost every computer system.
- ⇨ An Operating System is a collection of system programs that together control the operations of a computer system.
- ⇨ The purpose of an operating system is to provide an environment in which a user can execute programs.
- ⇨ The primary goal of an operating system is to make the computer system convenient to use.
- ⇨ A secondary goal is to use the computer hardware in an efficient manner.

Some popular examples of operating system are-

UNIX, Linux, MS-DOS, MS-Windows, Windows/NT, Android, OS/2, MacOS, MVC and VM etc.

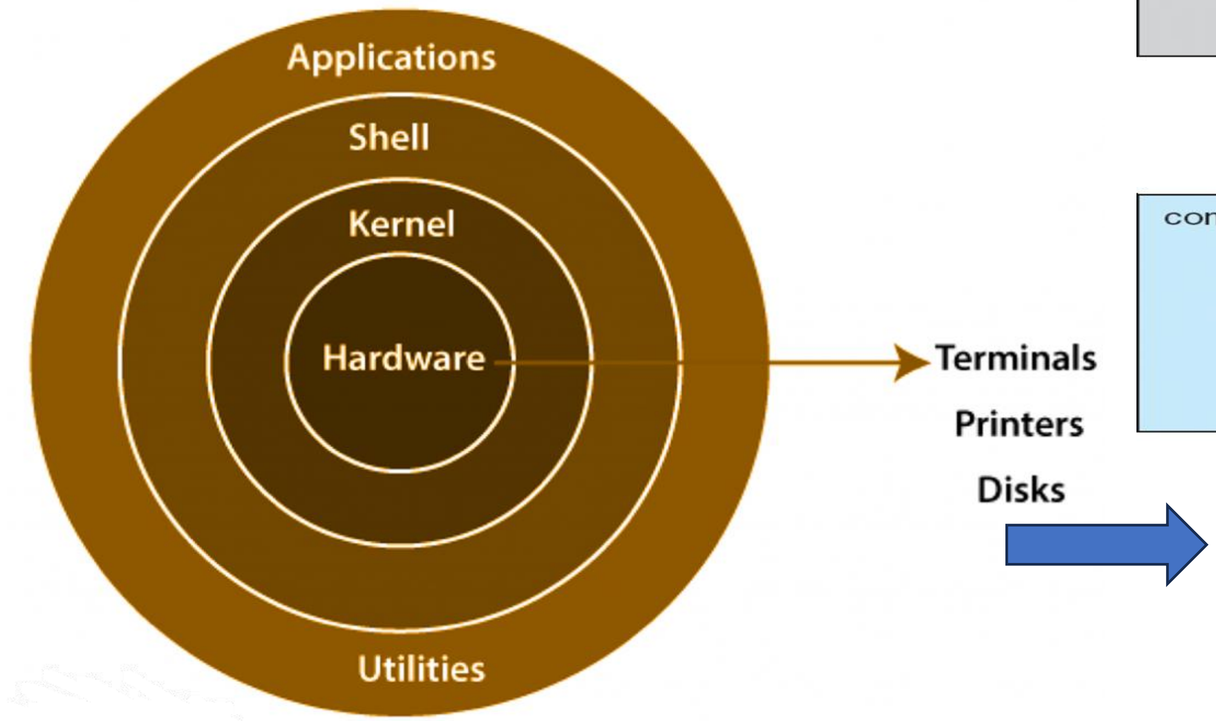
Operating system goals-

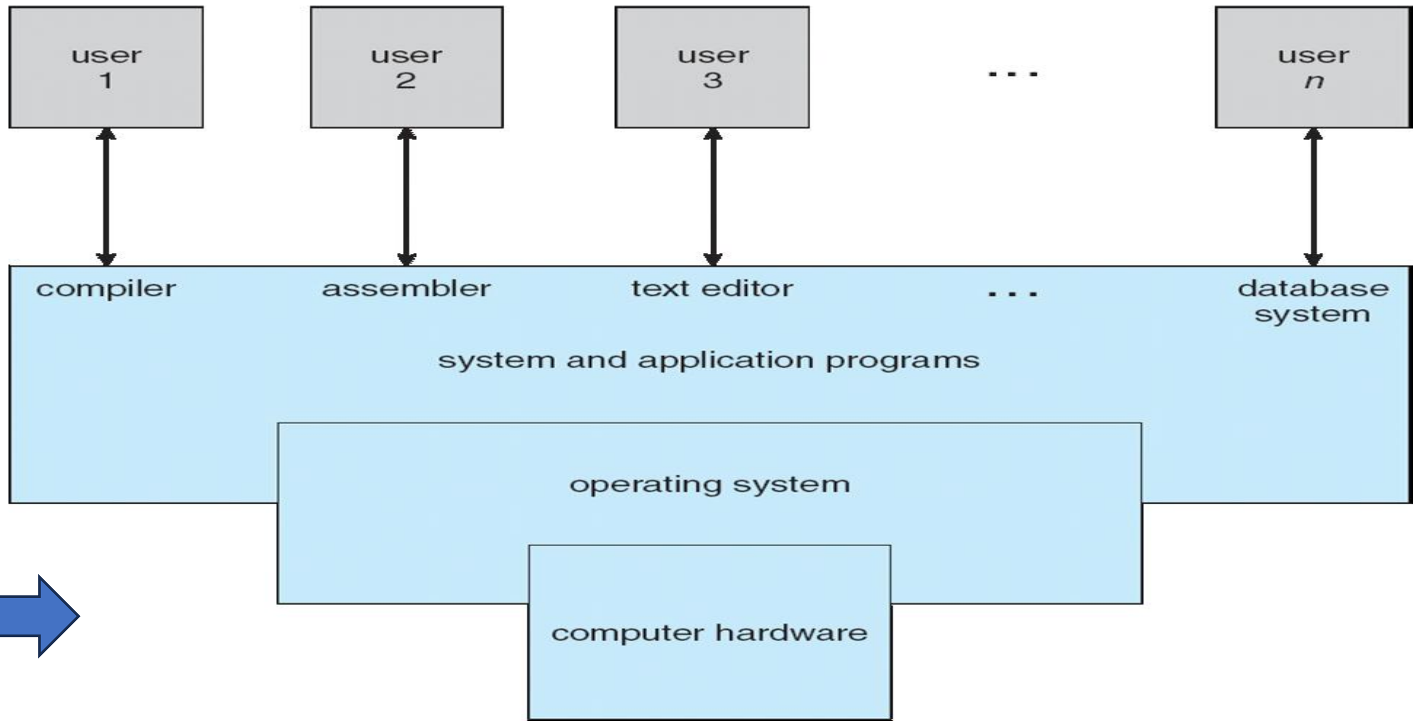
- Execute user programs and solve user problems easily.
- Make the computer system convenient to use.
- Use the computer hardware in an efficient manner.

Operating System Structure-

1. **Hardware –** provides basic computing resources like CPU, memory, I/O devices.
2. **Operating system –** controls and coordinates the use of the hardware among the various application programs for the various users.
3. **Applications programs** – Define the ways in which the system resources are used to solve the computing problems of the users (compilers, database systems, video games, business programs).
4. **Users-** (people, machines, other computers).

Layered Architecture of OS-





Abstract View of System Components

A system can be made modular in many ways .One method is layered approach, in which the OS is broken into number of layers where-

→The bottom layer(layer 0) is the hardware , the highest (layer N) is the user interface.

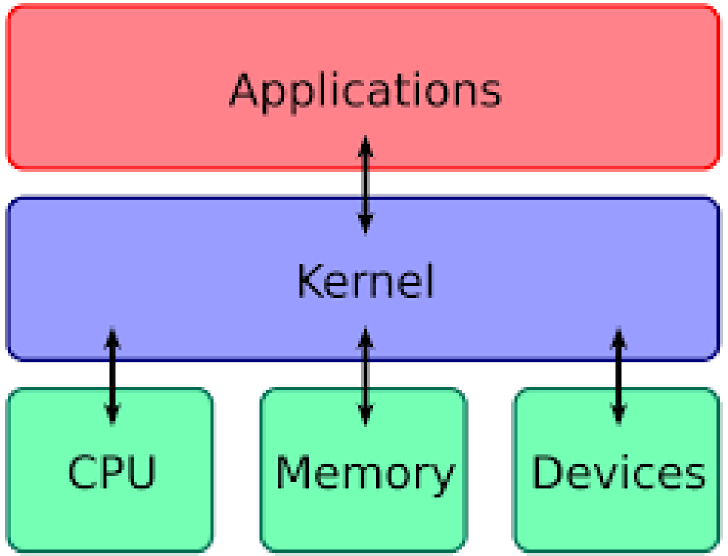
1.The outermost layer must be the User Interface layer.

2.The innermost layer must be the Hardware layer.

3.A particular layer can access all the layers present below it but it cannot access the layers present above it. That is layer n-1 can access all the layers from n-2 to 0 but it cannot access the nth layer.

Thus, if the user layer wants to interact with the hardware layer, the response will be traveled through all the layers from n-1 to 0. Each layer must be designed and implemented such that it will need only the services provided by the layers below it.

System Components:- Operating System has mainly two components-

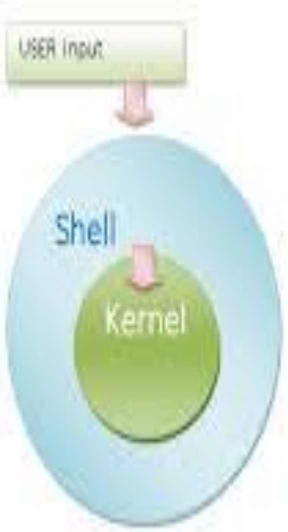


(1) Kernel-

- It is an active part of an Operating System.
- It is the part of OS running at all times.
- It is a program which can interact with the hardware. Ex: Device driver, .dll files, system files, devices, memory etc.

(2) Shell-

- It is called as the command interpreter.
- It is a set of programs used to interact with the application programs.
- It is responsible for execution of instructions given to OS (called commands).



Operating systems can be explored from two viewpoints:-

1. **User View:** From the user's point view, the OS is designed for one user to monopolize its resources, to maximize the work that the user is performing and for ease of use.

2. System View: From the computer's point of view, an operating system is a control program that manages the execution of user programs to prevent errors and improper use of the computer. It is concerned with the operation and control of I/O devices.

Lecture-2

Functions of Operating System

Functions/Features of Operating System:-

1. Process Management-

- A *process* is a **program in execution**.
- A process needs certain resources, including CPU time, memory, files, and I/O devices, to accomplish its task.

The operating system is responsible for the following activities in connection with process management.

✦ Process creation and deletion.
✦ process suspension and resumption.
✦ Provision of process mechanisms for:
- process synchronization (ensures the coordinated and orderly execution of multiple processes)
- process communication (ensures properly communication between multiple processes)

2. Main-Memory Management-

Memory is a large array of words or bytes, each with its own address. It is a repository of quickly accessible data shared by the CPU and I/O devices. Main memory is a volatile storage device. It loses its contents in the case of system failure.

The operating system is responsible for the following activities in connections with memory management:

♦ Keep track of which parts of memory are currently being used and by whom.
♦ Decide which processes to load when memory space becomes available.
♦ Allocate and de-allocate memory space as needed.

3. I/O System Management- The module that keeps track of the status of devices is called the I/O traffic controller. Each I/O device has a device handler that resides in a separate process associated with that device.

The I/O system consists of: -

✦ A memory management component that includes buffering, caching and spooling.
✦ A general device-driver interface for specific hardware devices.

4. File Management-

A file is a collection of related information defined by its creator. Commonly, files represent programs (both source and object forms) and data.

The operating system is responsible for the following activities in connections with file management: -

✦ File creation and deletion.
✦ Directory creation and deletion.
✦ Support of primitives for manipulating files and directories.
✦ Mapping files onto secondary storage.
✦ File backup on stable (non-volatile) storage media.

5. Secondary-Storage Management-

Since main memory (*primary storage*) is volatile and too small to accommodate all data and programs permanently, the computer system must provide *secondary storage* to back up main memory.

Most modern computer systems use disks as the principle on-line storage medium, for both programs and data.

The operating system is responsible for the following activities in connection with disk management: -

✦ Free space management
✦ Storage allocation
✦ Disk scheduling

6. Protection System-

Protection refers to a mechanism for controlling access by programs, processes, or users to both system and user resources.

The protection mechanism must: -

✦ distinguish between authorized and unauthorized usage.
✦ specify the controls to be imposed.
✦ provide a means of enforcement.

7. Networking (Distributed Systems)-

A distributed system is a collection processors that do not share memory or a clock. Each processor has its own local memory. The processors in the system are connected through a communication network. Communication takes place using a protocol. A distributed system provides user access to various system resources.

Access to a shared resource allows following:-

1. Computation speed-up
2. Increased data availability
3. Enhanced reliability

8. Command-Interpreter System- Many commands are given to the operating system by control statements which deal with:

✦ process creation and management
✦ I/O handling
✦ secondary-storage management
✦ main-memory management
✦ file-system access
✦ protection
✦ networking
✦ command-line interpreter
✦ shell (in UNIX)

- Its function is to get and execute the next command statement.

History of Operating System:-Following table shows the history of Operating Systems.

Generation	Year	Electronic devices used	Types of OS
First	1945 – 55	Vacuum tubes	Plug boards
Second	1955 – 1965	Transistors	Batch system
Third	1965 – 1980	Integrated Circuit (IC)	Multiprogramming
Fourth	Since 1980	Large scale integration	Unix, MS-DOS

Assembler:-

- Input to an assembler is an assembly language program.

Compilers:-

- A compilers is a program that accepts a source program in a high-level language and produces a corresponding object program.

Loader:-

- A loader is a routine that loads an object program and prepares it for execution. There are various loading schemes: absolute, relocating and direct-linking.
- In general, the loader must load, relocate and link the object program.

Lecture-3

Operating System Services:-

- The operating system service is a specific background process or set of special programs that run on a computer system which allows it to function properly.
- It controls input-output devices, execution of programs, managing files, etc.

Services of Operating System are as follows-

S.No	Services	S.No	Services
1.	Program Execution	2.	Input Output Operations
3.	Communication between Process	4.	File Management
5.	Memory Management	6.	Process Management
7.	Security and Privacy	8.	Resource Management
9.	User Interface	10.	Networking
11.	Error Handling	12.	Time Management

1. **Program Execution-** It is the Operating System that manages how a program is going to be executed. It loads the program into the memory after which it is executed. The order in which they are executed depends on the CPU Scheduling Algorithms like FCFS, SJF (Shortest Job First), etc. When the program is in execution, the Operating System also handles deadlock i.e. no two processes come for execution at the same time. The Operating System is responsible for the smooth execution of both user and system programs. The Operating System utilizes various resources available for the efficient running of all types of functionalities.

2. **Input Output Operations-** Operating System manages the input-output operations and establishes communication between the user and device drivers. Device drivers are software that is associated with hardware that is being managed by the OS so that the sync between the devices works properly. It also provides access to input-output devices to a program when needed.

3. **Communication Between Processes-** The Operating system manages the communication between processes. Communication between processes includes data transfer among them. If the processes are not on the same computer but connected through a computer network, then also their communication is managed by the Operating System itself.

4. **File Management-** The operating system helps in managing files also. If a program needs access to a file, it is the operating system that grants access. These permissions include read-only, read-write, etc. It also provides a platform for the user to create, and delete files. The Operating System is responsible for making decisions regarding the storage of all types of data or files, i.e, floppy disk/hard disk/pen drive, etc. The Operating System decides how the data should be manipulated and stored.

5. Memory Management- Let's understand memory management by OS in simple way. Imagine a cricket team with limited number of player . The team manager (OS) decide whether the upcoming player will be in playing 11 ,playing 15 or will not be included in team , based on his performance . In the same way, OS first check whether the upcoming program fulfil all requirement to get memory space or not ,if all things good, it checks how much memory space will be sufficient for program and then load the program into memory at certain location. And thus , it prevents program from using unnecessary memory.

6. Process Management- Let's understand the process management in unique way. Imagine, our kitchen stove as the (CPU) where all cooking(execution) is really happened and chef as the (OS) who uses kitchen-stove(CPU) to cook different dishes(program). The chef(OS) has to cook different dishes(programs) so he ensure that any particular dish(program) does not take long time(unnecessary time) and all dishes(programs) gets a chance to cooked(execution) .The chef(OS) basically scheduled time for all dishes(programs) to run kitchen(all the system) smoothly and thus cooked(execute) all the different dishes(programs) efficiently.

7. Security and Privacy-

- **Security :** OS keep our computer safe from an unauthorized user by adding security layer to it. Basically, Security is nothing but just a layer of protection which protect computer from bad guys like viruses and hackers. OS provide us defenses like firewalls and anti-virus software and ensure good safety of computer and personal information.
- **Privacy :** OS give us facility to keep our essential information hidden like having a lock on our door, where only you can enter and other are not allowed . Basically , it respect our secrets and provide us facility to keep it safe.

8. Resource Management- System resources are shared between various processes. It is the Operating system that manages resource sharing. It also manages the CPU time among processes using CPU Scheduling Algorithms. It also helps in the memory management of the system. It also controls input-output devices. The OS also ensures the proper use of all the resources available by deciding which resource to be used by whom.

9. User Interface- User interface is essential and all operating systems provide it. Users either interacts with the operating system through the command-line interface or graphical user interface or GUI. The command interpreter executes the next user-specified command. A GUI offers the user a mouse-based window and menu system as an interface.

10. Networking- This service enables communication between devices on a network, such as connecting to the internet, sending and receiving data packets, and managing network connections.

11. Error Handling- The Operating System also handles the error occurring in the CPU, in Input-Output devices, etc. It also ensures that an error does not occur frequently and fixes the errors. It also prevents the process from coming to a deadlock. It also looks for any type of error or bugs that can occur while any task. The well-secured OS sometimes also acts as a countermeasure for preventing any sort of breach of the Computer System from any external source and probably handling them.

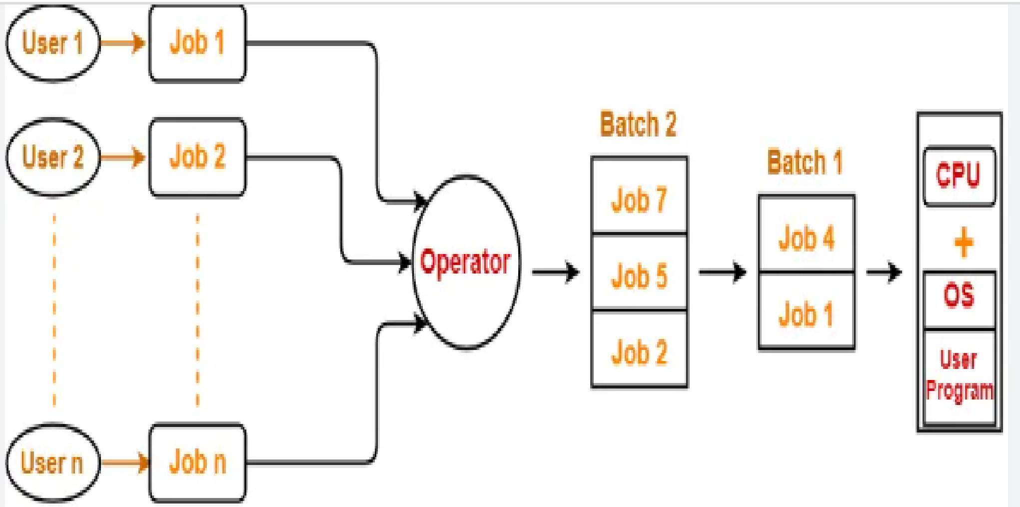
12. Time Management- Imagine traffic light as (OS), which indicates all the cars(programs) whether it should be stop(red)=>(simple queue), start(yellow)=>(ready queue),move(green)=>(under execution) and this light (control) changes after a certain interval of time at each side of the road(computer system) so that the cars(program) from all side of road move smoothly without traffic.

Classification/Types of Operating Systems-

- Operating system may be classified in different ways according to their uses and number of processors in computer system.
- Some computer systems have single processor and others have multiprocessors. So based on the processors used in computer systems, operating systems are classified into the following categories:-

1. Batch Processing Operating System:-

- This type of OS accepts more than one jobs/processes and these jobs are batched/ grouped together according to their **similar requirements.**
- To speed up processing, OS batched the jobs with similar needs and run them together as a group.
- This is done by computer operating system. Whenever the CPU becomes available, the batched jobs are sent for execution and gradually the output is sent back to the user.
- It allowed only one program at a time.
- This OS is responsible for scheduling the jobs according to priority, types and the resource required.
- One of the key benefits of a Batch Operating System is that **it can handle multiple users simultaneously.**



- The disadvantage of batch system is that the CPU is often idle because users need to wait for their turn to use the resources.
- Sometimes It can be difficult to manage multiple users. If one user is not done with their work, it can hold up the entire system.

2. Time-Sharing Operating Systems:-

Time sharing OS is a logical extension of multiprogramming. It provides extra facilities such as:

- Faster switching between multiple jobs to make processing faster.
- Allows multiple users to share computer system simultaneously.
- The users can interact with each job while it is running.

These systems use a concept of virtual memory for effective utilization of memory space. Hence, in this OS, no jobs are discarded. It uses CPU scheduling, memory management, disc management and security management.

Examples: **CTSS, MULTICS, CAL, UNIX etc**.

3. Real-Time Operating Systems (RTOS):-

- A real-time operating system (RTOS) is a multitasking operating system intended for applications with fixed deadlines (real-time computing).
- Such applications include some small embedded systems, automobile engine controllers, industrial robots, spacecraft, industrial control, and some large-scale computing systems.
- The real time operating system can be classified into two categories:

1. Hard Real Time Operating System :- In Hard RTOS, all critical tasks must be completed within the specified time duration, i.e., within the given deadline. Not meeting the deadline would result in critical failures such as damage to equipment or even loss of human life. Example- **Airbags provided by carmakers**, **Air Traffic Control system**, **Missile control system** etc.

2. Soft Real Time Operating System:- Soft RTOS accepts a few delays via the means of the Operating system. In this kind of RTOS, there may be a closing date assigned for a particular job, but a delay for a small amount of time is acceptable. So, cut off dates are treated softly via means of this kind of RTOS.

For Example, This type of system is used in **Online Transaction systems** and **Livestock price quotation systems**.

"A Real Time OS(RTOS) has well defined fixed time constraints. Processing must be done within the time constraint or the system will fail. A real time system is said to function correctly only if it returns the correct result within the time constraint. These systems are characterized by having time as a key parameter."

4. Multiprocessor Operating Systems:-

- Multiprocessor operating systems are also known as parallel OS or tightly coupled OS.
- Such operating systems have more than one processor in close communication that share the computer bus, the clock and sometimes memory and peripheral devices.
- It executes multiple jobs at same time and makes the processing faster.

Multiprocessor systems have three main advantages:-

➲ **Increased throughput:** By increasing the number of processors, the system performs more work in less time. The speed-up ratio with **N** processors is less than **n**.

➲ **Economy of scale:** Multiprocessor systems can save more money than multiple single-processor systems, because they can share peripherals, mass storage, and power supplies.

➲ **Increased reliability:** If one processor fails to done its task, then each of the remaining processors must pick up a share of the work of the failed processor. The failure of one processor will not halt the system, only slow it down.

The multiprocessor operating systems are classified into two categories:-

➲ **In symmetric multiprocessing system (SMP),** each processor runs an identical copy of the operating system, and these copies communicate with one another as needed.

➲ **In asymmetric multiprocessing system (AMS),** a processor is called master processor that controls other processors called slave processor. Thus, it establishes master-slave relationship. The master processor schedules the jobs and manages the memory for entire system. It's called as **Master – Slave Processors.**

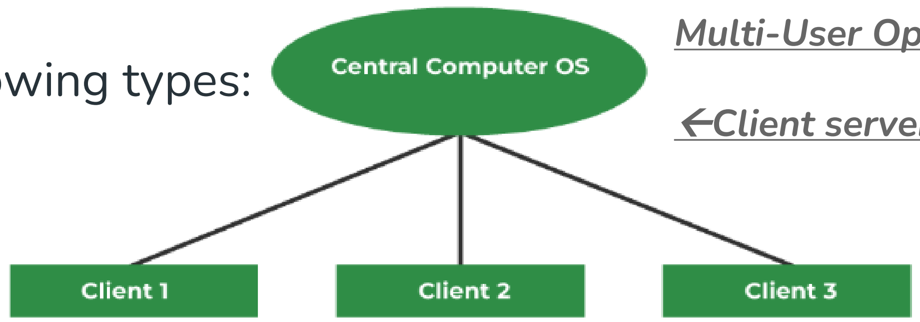
Lecture-4

5. Multiuser Systems-

- In a multiuser operating system, multiple numbers of users can access different resources of a computer at the same time.
- The access is provided using a network that consists of various personal computers attached to a mainframe computer system(called as server).
- A multi-user operating system allows the permission of multiple users for accessing a single machine at a time. The various personal computers can send and receive information to the mainframe computer system. Thus, the mainframe computer acts as the server and other personal computers act as clients for that server.

Types of Multi-user Operating Systems: -

The multi-user operating systems is of the following types:



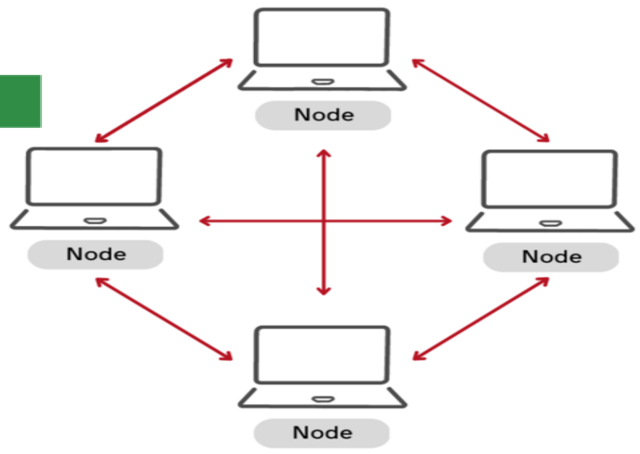
Multi-User Operating System

←Client server system

a) Distributed System
b) Time sliced system
c) Multiprocessor system

a) Distributed system:-

- The distributed operating system also known as distributed computing is a collection of multiple computers located on different locations.
- This OS communicate and coordinate their actions by passing messages to one another from any system.
- These all systems emulate a single coherent system to the end user. The end users will communicate with them with the help of the network.
- This system is divided in a way that multiple requests can be handled and in turn, the individual request can be satisfied eventually.
- **Examples:** E-Banking, Mobile apps



Peer-to-Peer systems

It can be classified into two categories:

1. Client-Server systems
2. Peer-to-Peer systems

b) Time-sliced system:-

- It is the system **where each user is allocated to a short span of CPU time**.
- CPU time is divided into small time slices, and one time slice is assigned to each task.
- The decision to run the next piece of the job is decided by the schedule. This schedule executes the run instructions that need to be executed.
- The user can take turns and thus the operating system will handle the user's request among the users who are connected.
- ***"This feature is not applicable in the single-user operating system."*** They use the mainframe system concurrently.

Example: Mainframe system

c) Multiprocessor system:-

It involves multiple processors at a time. Enhance the overall performance. If one processor fails other continues working. **Example:** Spreadsheets, Music player

Features:

The multi-processor operating system has the following features:

•Resource sharing: This maps to time slicing, multiple peripherals such as printers can be shared, different files or data can also be shared.
•Time-sharing
•Background sharing
•Invisibility: Many functions of multi-user operating systems are invisible to the user.

6. Interactive Operating System:-

- Interactive operating systems are computers that accept human inputs.
- Users give commands or some data to the computers by typing or by gestures. They facilitate interactive behavior. Mac, Windows, UNIX and Linux OS are some examples of interactive operating systems.

Some other definitions of Interactive OS:-

- ❖ An interactive operating system is an operating syste that allows the execution of interactive programs. All PC operating systems are interactive operating systems only.
- ❖ An interactive operating system gives permission to the user to interact directly with the computer. In an Interactive operating system, the user enters some command into the system and the work of the system is to execute it.

The advantages are as follows:-

1.Usability: An operating system is designed to perform something and the interactiveness allows the user to manage the tasks more or less in real-time.

2.Security: Simple security policy enhancement. In interactive systems, the user virtually always knows what their programs will do during their lifetime, thus allowing us to forecast and correct the bugs.

Disadvantages of Interactive Operating System:-

- **Tough to design:** Depending on the target device, interactivity might be proved challenging to design because the user must be prepared for every input.

7. Multi-Process System:-

- Multiprocessing is a system in which multiple processes can be run on single or multiple processors.
- In this, CPUs are added for increasing computing speed of the system because in multiprocessing, there are many processes that are executed simultaneously. Each process has its own set of resource (like CPU, memory etc).
- Each process can have its own scheduling algorithm.
- Multiprocessing is further classified into two categories: Symmetric Multiprocessing **(SMP)** and Asymmetric Multiprocessing (**Master – Slave Processors).**

Advantages of Multiprocessing:-

•**True Parallelism:** In Multiprocessing, tasks are executed in parallel across different processors, thus making task completion faster.

•**Increased Reliability:** In case one processor fails to work, the other processors are still able to work and this improves on the fault tolerance of the system.

•**High Performance:** Through multiprocessing where there are many processors performing a single task, systems can support complex applications and large databases thus increasing system throughput.

Disadvantages of Multiprocessing:-

•**High Cost:** Multiprocessing systems are generally costlier because of the inclusion of another or other hardware(s) (processors).

•**Complexity in System Design:** Multiprocessing adds the factor of synchronization of the processors, load balancing across the processors and communication between each of the processors.

•**Power Consumption:** Parallel processing requires multiple processors to be run simultaneously, thus a disadvantage in the use of power particularly for portable devices or those that are sensitive to power consumption.

Lecture-5

8. Multi-threaded System:-

- ✓ Multithreading is an operating system (OS) feature that allows multiple threads of a process to run concurrently.
- ✓ This improves performance and responsiveness and allows more than one user to access a program at the same time.
- ✓ Multithreading works by allowing a computer's processor to handle multiple tasks at the same time.
- ✓ Even though the processor can only do one thing at a time, it switches between different threads from various programs so quickly that it looks like everything is happening all at once.

Advantages:-

•**Processor Handling** : The processor can execute only one instruction at a time, but **it switches between different threads so fast that it gives the illusion of simultaneous execution.**

•**Thread Synchronization** : Each thread is like a separate task within a program. They share resources and work together smoothly, ensuring programs run efficiently.

•**Efficient Execution** : Threads in a program can run independently or wait for their turn to process, making programs faster and more responsive.

•**Programming Considerations** : Programmers need to be careful about managing threads to avoid problems like conflicts or situations where threads get stuck waiting for each other.

Process Vs Thread

S.NO	Process	Thread
1.	Process means any program is in execution.	Thread means a segment of a process.
2.	The process takes more time to terminate.	The thread takes less time to terminate.
3.	It takes more time for creation.	It takes less time for creation.
4.	It also takes more time for context switching.	It takes less time for context switching.
5.	The process is less efficient in terms of communication.	Thread is more efficient in terms of communication.
6.	Multiprogramming holds the concepts of multi-process.	We don't need multi programs in action for multiple threads because a single process consists of multiple threads.

Difference between Multiprogramming, Multitasking, Multithreading & Multiprocessing

1.Multiprogramming – Multiprogramming is known as keeping multiple programs in the main memory at the same time ready for execution.

2.Multiprocessing – A computer using more than one CPU at a time.

3.Multitasking – Multitasking is nothing but is multiprogramming with a Round-robin scheduling algorithm.

4.Multithreading is an extension of multitasking.

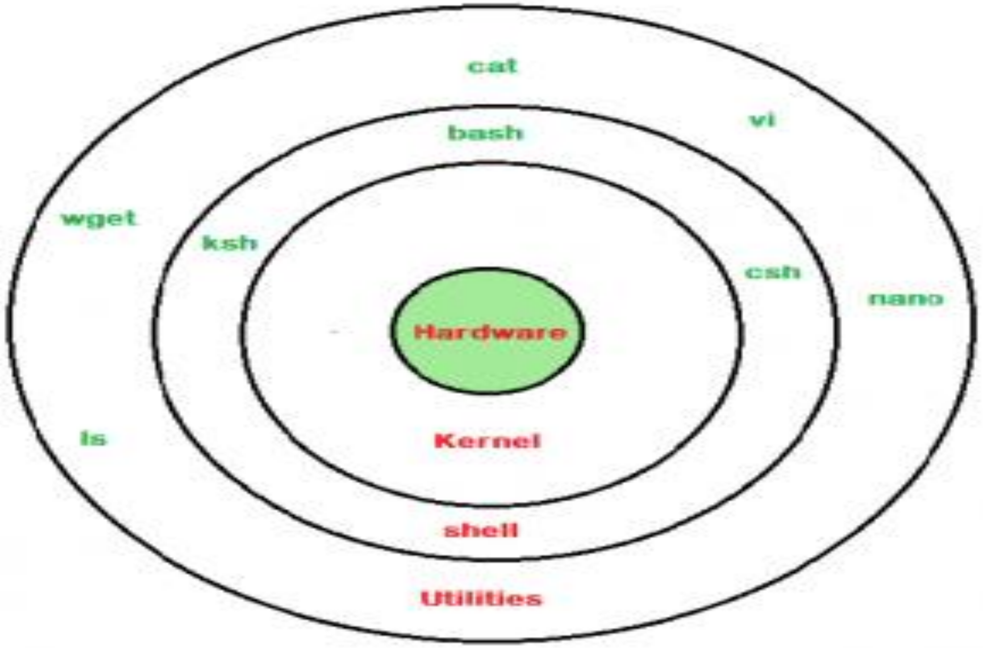
Feature	Multiprogramming	Multitasking	Multithreading	Multiprocessing
Definition	Running multiple programs on a single CPU	Running multiple tasks (applications) on a single CPU	Running multiple threads within a single task (application)	Running multiple processes on multiple CPUs (or cores)
Resource Sharing	Resources (CPU, memory) are shared among programs	Resources (CPU, memory) are shared among tasks	Resources (CPU, memory) are shared among threads	Each process has its own set of resources (CPU, memory)
Scheduling	Uses round-robin or priority-based scheduling to allocate CPU time to programs	Uses priority-based or time-slicing scheduling to allocate CPU time to tasks	Uses priority-based or time-slicing scheduling to allocate CPU time to threads	Each process can have its own scheduling algorithm
Memory Management	Each program has its own memory space	Each task has its own memory space	Threads share memory space within a task	Each process has its own memory space
Context Switching	Requires a context switch to switch between programs	Requires a context switch to switch between tasks	Requires a context switch to switch between threads	Requires a context switch to switch between processes
Inter-Process Communication (IPC)	Uses message passing or shared memory for IPC	Uses message passing or shared memory for IPC	Uses thread synchronization mechanisms (e.g., locks, semaphores) for IPC	Uses inter-process communication mechanisms (e.g., pipes, sockets) for IPC

Lecture-6

What is Shell?

- It is called as the command interpreter. It is a set of programs used to interact with the application programs.
- It is responsible for execution of instructions given to OS (called commands).
- A shell is a special user program that provides an interface for the user to use operating system services.
- Shell accepts human-readable commands from users and converts them into something which the kernel can understand.
- It is a command language interpreter that executes commands read from input devices such as keyboards or from files.
- The shell gets started when the user logs in or starts the terminal.

Shell is broadly classified into two categories –



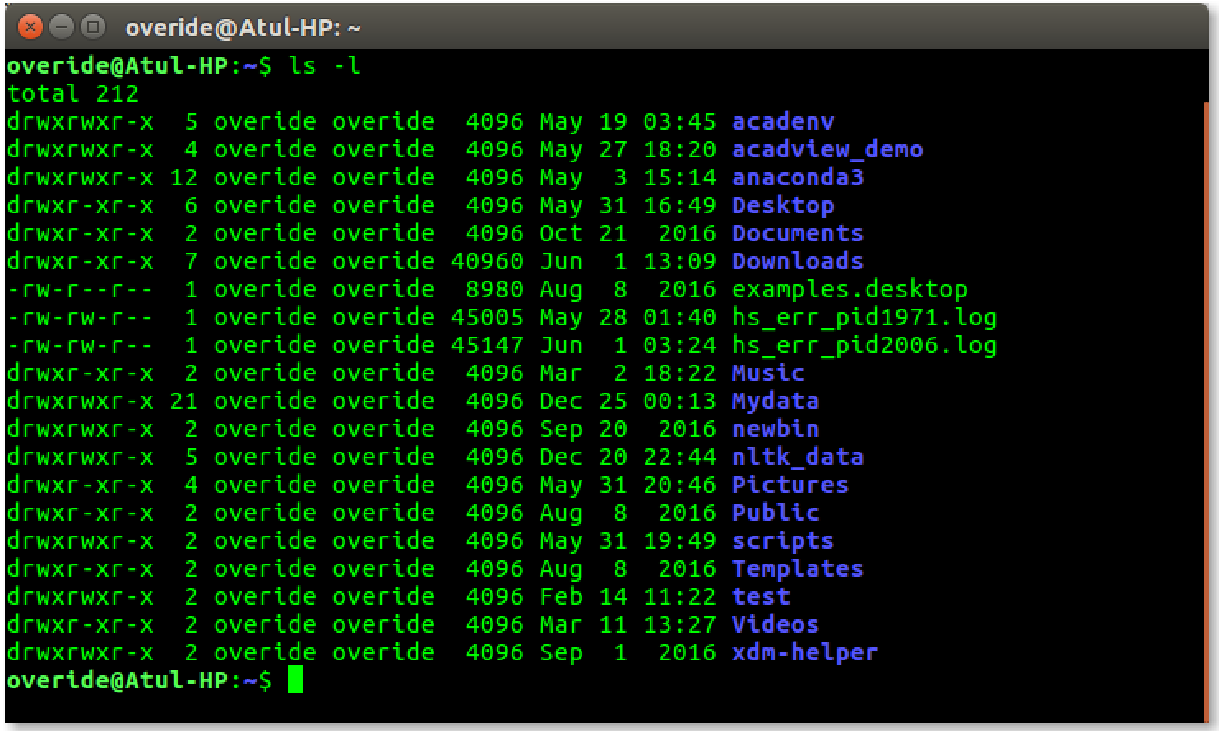
Command Line Shell:-

1. Shell can be accessed by users using a command line interface.
2. A special program called Terminal in Linux/Mac OS, or Command Prompt in Windows OS is provided to type in the human-readable commands such as "cat", "ls" etc. and then it is being executed.
3. The result is then displayed on the terminal to the user.

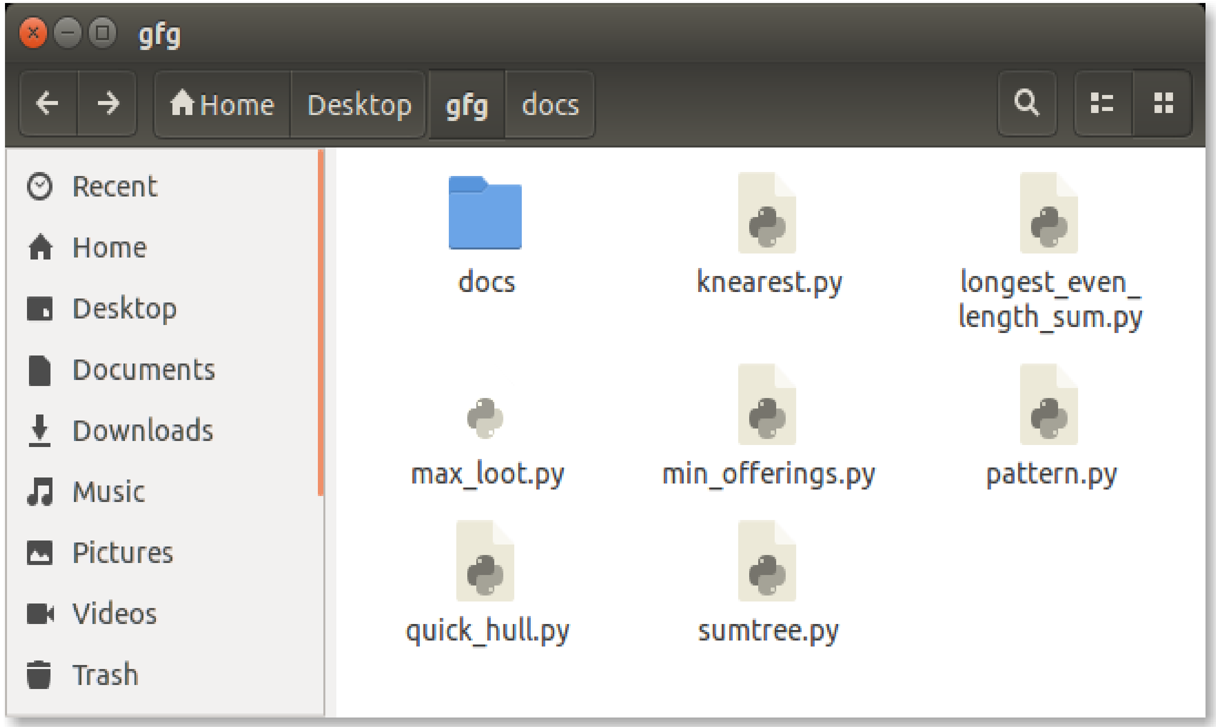
Graphical Shell:-

Graphical shells provide means for manipulating programs based on the graphical user interface (GUI), by allowing for operations such as opening, closing, moving, and resizing windows, as well as switching focus between windows.

Window OS or Ubuntu OS can be considered as a good example which provides GUI to the user for interacting with the program. Users do not need to type in commands for every action.



Linux command line



GUI Shell in Ubuntu system

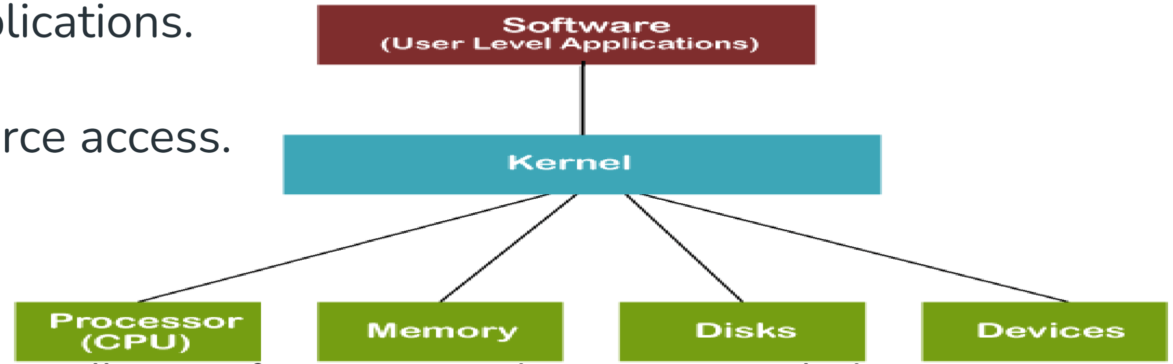
Lecture-7

What is Kernel?

- ❑ It acts as a bridge/interface between software applications and the hardware of a computer.
- ❑ The kernel manages the system's resources and facilitates communication between hardware and software components.
- ❑ "***Kernel is a computer program that is a core or heart of an operating system with complete control over everything in the system.***"

It manages the following resources of the operating system –

1. It manages system resources like CPU, memory and devices. It ensures everything works together smoothly and efficiently.
2. It handles tasks like running programs, accessing files and connecting to devices like printers and keyboards.
3. Facilitates communication between hardware and user applications.
4. Ensures efficient and secure multitasking.
5. Manages system stability and prevents unauthorized resource access.



Types of Kernel:- The kernel has different types-

1. **Monolithic Kernel-**

- ❑ A monolithic kernel is an operating system architecture where all core functions and services, including memory management, process scheduling, device drivers and file systems reside in a single, large kernel space. It means they all run in the same memory space.
- ❑ It has dependencies between systems components.
- ❑ It has huge lines of code which is complex.

Examples of Monolithic Kernels are **Unix, Linux, Open VMS, XTS-400, etc.**

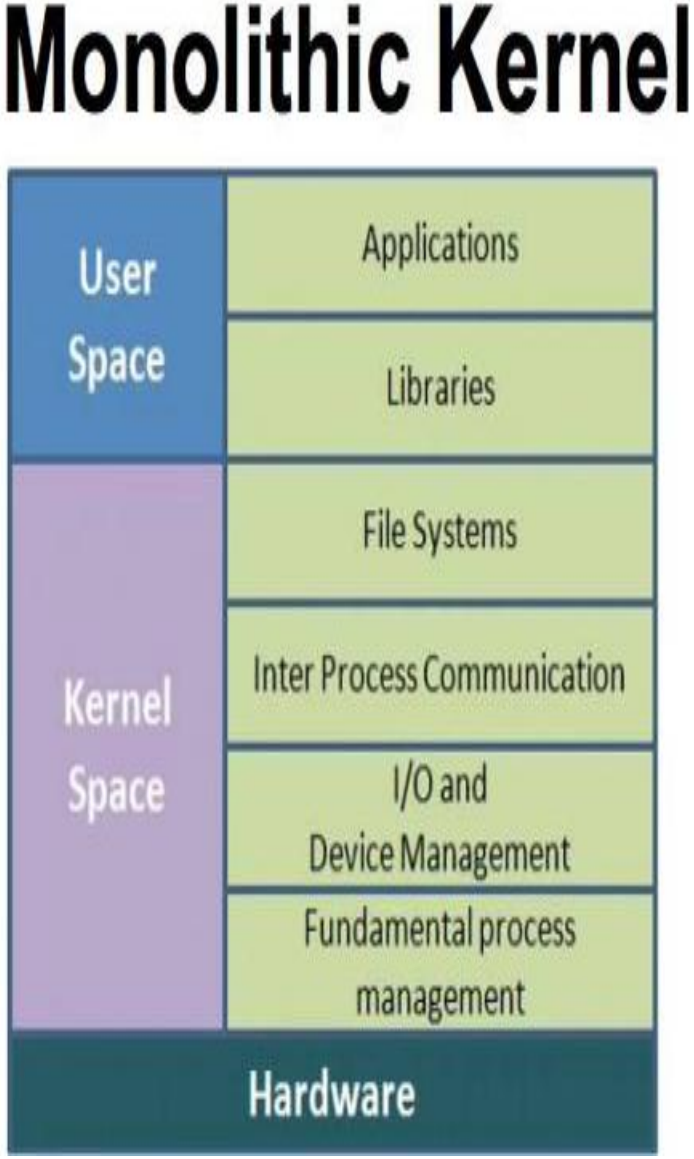
Advantages:-

1. **Efficiency:** Monolithic kernels are generally faster than other types of kernels because they don't have to switch between user and kernel modes for every system call.
2. **Tight Integration:** Since all the operating system services are running in kernel space, they can communicate more efficiently with each other, making it easier to implement complex functionalities and optimizations.

3. Simplicity: Monolithic kernels are simpler to design, implement, and debug than other types of kernels because they have a unified structure that makes it easy to manage the code.

4. Lower latency: Monolithic kernels have lower latency (delay) than other types of kernels because system calls and interrupts can be handled directly by the kernel.

5. Direct Interaction: Kernel services can directly interact with each other without needing complex inter-process communication (IPC) mechanisms. Also Kernel services have direct access to hardware resources, which can be advantageous for performance-critical tasks.



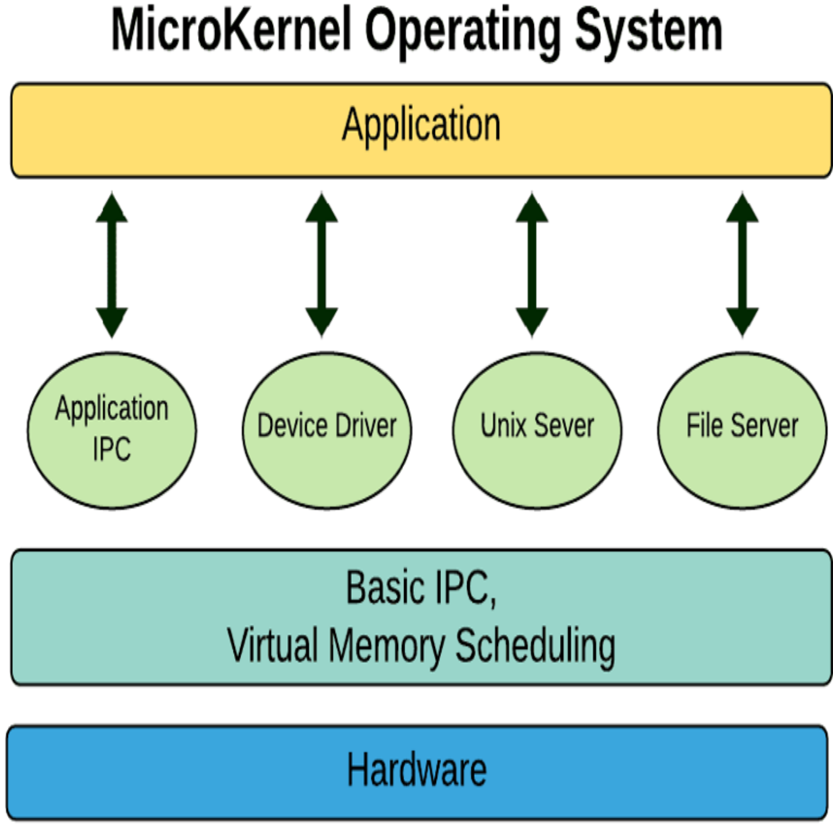
Disadvantages:-

1. Stability Issues: Monolithic kernels can be less stable than other types of kernels because any bug or security vulnerability in a kernel service can affect the entire system.

2. Maintenance Difficulties: Monolithic kernels can be more difficult to maintain than other types of kernels because any change in one of the services can affect the entire system.

3. Limited Modularity: Monolithic kernels are less modular than other types of kernels because all the operating system services are tightly integrated into the kernel space. This makes it harder to add or remove functionality without affecting the entire system.

2. Microkernel System:-



- A Microkernel is an approach to design an Operating System (OS) and provides very fundamental services required to run the OS.
- In this kernel, **all non essential components are moved from kernel space to user space.**
- Microkernels provide a minimalist approach to operating system design by running **only essential services like basic memory management, task scheduling, IPC** etc. in kernel space which makes the OS more modular and secure.
- Microkernels use **Inter-Process Communication (IPC)** for communication.

Features of Microkernel-Based Operating System:-

•**Minimal Core :** The kernel only manages basic tasks like CPU scheduling, memory management, and inter-process communication. All other services (like device drivers) run in user space, outside the kernel.

•**Modularity :** System services are separated into independent modules. This makes it easy to update or replace individual components without affecting the entire system.

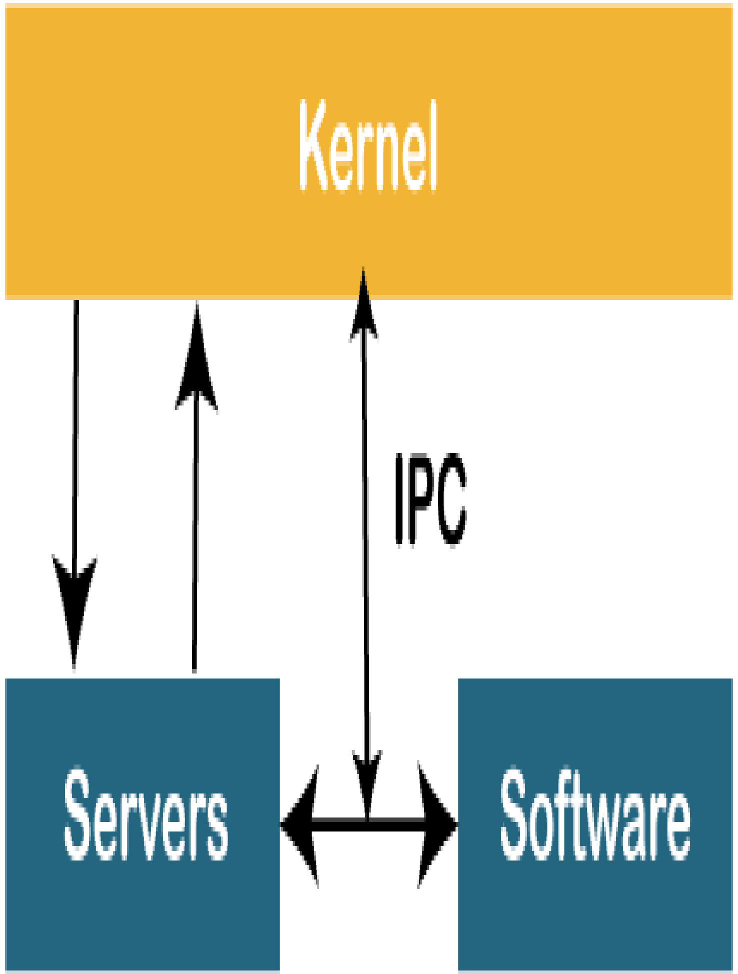
•**Stability and Fault Isolation :** If a service like a device driver fails, it won't crash the whole system. Each service runs independently, so the core OS remains stable.

•**Ease of Maintenance :** Since services are independent, fixing bugs or adding new features doesn't require major changes to the kernel itself, making maintenance easier.

Examples of Microkernel are **L4, AmigaOS, Minix, K42**, etc.

Advantages of Microkernel:-

- **Modularity:** The microkernel design allows greater modularity. This can make adding and removing features and services from the system easier.
- **Performance:** Because the kernel only contains the essential functions required to manage the system, the microkernel design can improve performance. This can make the system faster and more efficient.
- **Security:** The microkernel design can improve security by reducing the system's attack surface by limiting the functions provided by the kernel. Malicious software may find it more difficult to compromise the system as a result of this.
- **Reliability:** Microkernels are less complex than monolithic kernels, which can make them more reliable and less prone to crashes or other issues.
- **Scalability:** Microkernels can be easily scaled to support different hardware architectures, making them more versatile.
- **Portability:** Microkernels can be ported to different platforms with minimal effort, which makes them useful for embedded systems and other specialized applications.



Disadvantages of a Microkernel:-

- **Slower Message Passing:** Slower message passing between user-level processes can affect performance, especially in high-performance applications. Smaller in size but slower in execution.
- **More Complex:** Increased complexity due to the modular design can make it more difficult to develop and maintain the operating system .
- **Limited Performance:** Limited performance optimization due to separation of kernel and user-level processes.
- **Higher Memory Usage:** Higher memory usage compared to a monolithic kernel.

Lecture-8

3. Reentrant Kernel:-

1. A reentrant kernel in an operating system which allows multiple processes or threads to run in kernel mode simultaneously without affecting one another.
2. In kernel mode, a reentrant kernel allows processes to give up the CPU. They have no effect on other processes entering or running in kernel mode.
3. Reentrant kernel is reusable and can be interrupted and invoked multiple times simultaneously.
4. If a computer program or routine can be safely called again and again before its previous invocation has been completed, it is said to be reentrant (i.e it can be safely executed concurrently).

Why is reentrant kernel useful?

- Reentrant kernel is a desirable feature for multithreaded programs, allowing multiple threads to execute the function concurrently without any issues.
- A reentrant kernel provides better performance because there is no contention for the kernel. A kernel that is not reentrant needs to use a lock to make sure that no two processes are executing kernel code at the same time.
- A reentrant function can be interrupted at any time and resumed at a later time without loss of data.
- Reentrant kernel modify local data not any global data.
- This is crucial for modern operating systems to maintain responsiveness and handle events like I/O completion and interrupts without blocking other processes.
- It must only work with the data provided by the caller.
- A reentrant kernel enables a scheduler to switch between different kernel control paths while one is still in progress, allowing multiple processes to access the kernel concurrently.

- **Handling Events-**

When a process in kernel mode is waiting for an I/O operation or an interrupt, the reentrant kernel can switch to another process, preventing the system from becoming idle or unresponsive.

Example:

Imagine a process waiting for data from a disk. With a reentrant kernel, the scheduler can switch to another process while the disk operation is ongoing. When the data is ready, the original process can be resumed.

Advantages of Reentrant kernel:-

1. Improved System Responsiveness

- **No Contention for the Kernel**
- **CPU Time Saving**
- **Interrupt Handling**

2. CPU Efficiency

- **Better Resource Utilization**

3. Flexibility and Reliability

- **Concurrency**